

$$\hat{\lambda}_{5t} = \left(\frac{5+6+8+9+7}{5} \right) \% \approx 7\%$$

Problem Set 1 Financial Market and Expectation (Solution)

1. If the expected path of one-year interest rates over the next five years is 5 percent, 6 percent, 8 percent, 9 percent, 7 percent, then what is the today's interest rate on the five-year bond?

$$\hat{\lambda}_5 = \frac{5\% + 6\% + 8\% + 9\% + 7\%}{5} \approx 7\%$$

$$\hat{\lambda}_5 = \frac{5+6+8+9+7}{5} \% \approx 7\%$$

2. Consider 1-year zero-coupon bond and 2-year zero-coupon bond. Both bonds have the same face value of \$1. Today's 1-year interest rate on the 1-year bond is 10%. Investors are risk-neutral.

$$\hat{\lambda}_{2t} = \left(\frac{10+10}{2} \right) \% = 10\%$$

$$\hat{\lambda}_{1t+1} = 10\%$$

(a) Suppose that investors expect that the 1-year interest rate on the 1-year bond next year will be 10%. There is no uncertainty, meaning that investors assume that the 1-year interest rate next year will be exactly 10%. Given these information, what would be today's 2-year interest rate on the 2-year bond?

$$\hat{\lambda}_{2t} = \frac{1}{2} (\hat{\lambda}_{1t} + \hat{\lambda}_{1t+1}^e) = 10\%$$

(b) Now the expected rate is still 10% but suppose there is uncertainty: with probability 0.5 the 1-year interest rate on the 1-year bond next year will be 12%, and with probability 0.5 the 1-year interest rate on the 1-year bond next year will be 8%.

What would be today's 2-year interest rate on the 2-year bond in this case?

$$\hat{\lambda}_{2t} = \frac{1}{2} \left(10\% + \frac{1}{2} 12\% + \frac{1}{2} 8\% \right) = 10\%$$

(c) Continue from (b) above, but now suppose that the investors are risk-averse. In this case, today's 2-year interest rate will be higher or lower than your answer in (b)? Why?

(a)

$$i = \frac{10\% + 10\%}{2} = 10\%$$

(b)

$$i = \frac{10\% + 10\%}{2} = 10\%$$

(c) If the investor is risk-averse, even if the expected interest rate remains at 10% in the second year, they will ask a risk premium due to the existence of risk. Assume

that the risk premium is denoted as x , then $\hat{\lambda}_{2t} = \frac{1}{2}(\hat{\lambda}_{1t} + \hat{\lambda}_{1t+1}^e + x) > 10\%$

$$i = \frac{10\% + (10\% + x)}{2} > 10\%$$

Therefore, the investor will ask a higher interest rate than in part (b) to compensate for the utility loss resulting from the risk they undertake.

3. Suppose you are to inherit \$10,000 a year from now and have just sold your rights to that inheritance for \$9,700, and that you would have accepted no less than \$9,389 in exchange for those rights. What is the current interest rate?

$$9,700 = \frac{10,000}{1+r} \Rightarrow r = 3.09\%$$

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FV = face value

4. A Company issues consol with a coupon rate of 10% and a face value of \$100,000. If the yield to maturity is 8%, what is the issue price of the bond?

Let z be the constant payment, P_c be the price of the bond, then

$$\text{issue price} = \frac{\text{Payment}}{\text{YTM}} \quad z = 100,000 \times 10\% = 10,000$$

$$\text{YTM} = \frac{\text{FV} - \text{PV}}{\text{PV}} \times 100\% \quad \boxed{P_c = \frac{z}{i} = \frac{10,000}{8\%} = 125,000}$$

$$100,000 \times 10\% = 10,000$$

$$\Rightarrow PV = 92,592.59$$

$$0.08 = \frac{100,000 - x}{x}$$

$$1.08x = 100,000$$

5. Suppose there is a perpetual bond that pays \$100,000 in annual interest. An investor purchases the bond on the day after the interest payment at a yield to maturity of 12% and holds it for one year (having received the interest). After one year, the investor sells the bond at a yield to maturity of 4%, assuming that the inflation rate π_t is 20% during that year.

(a) What is the nominal rate of return (i) for the investor?

(b) What is the real rate of return r

Let P_t be the price that the investor buy from company, P_{t+1} be the price that the investor sell to someone, C_{t+1} be the interest revenue.

$$P_t = \frac{100,000}{12\%} = 833,333.3$$

$$\text{normal rate of return: } \hat{\lambda} = \frac{P_{t+1} + C_{t+1} - P_t}{P_t} \times 100\%$$

real rate of return:

$$= \frac{2,500,000 + 100,000 - 833,333.3}{833,333}$$

$$r = \frac{1 + \hat{\lambda}}{1 + \pi_t} - 1 = \frac{1 + 212\%}{1 + 20\%} - 1 = 160\%$$

$$P_{t+1} = \frac{100,000}{4\%} = 2,500,000$$

$$C_{t+1} = 100,000$$

$$= 212\%$$

(a) $i = \frac{(P_{t+1} - P_t) + C_{t+1}}{P_t} \times 100\% = 212\%$

(b) $r = \frac{\frac{P_{t+1} + C_{t+1}}{1 + \pi} - P_t}{P_t} \times 100\% = 160\%$

or

$$r = \frac{1 + 212\%}{1 + 20\%} - 1 = 160\%$$

6. In bond markets, the term "yield to maturity, YTM" represent the average annual rate of returns available for investing in the bonds with the given maturity. The data suggest that the yield for 5-year Treasury (US government) bond is usually higher than the yields for the 3-year Treasury bond.

(a) Can you explain why the yield of the 5-year Treasury bond is usually higher than the 2-year Treasury bond?

(b) On December 3, 2018, the 5-year yield fell below the 3-year yield, which caused the Dow Jones Index to dropped by 800 points. Can you explain why the 5-year yield can be lower than the 3-year yield? What signals are conveyed by this yield inversion?

(a) When the yield on a 5-year bond is higher than the yield on a 2-year bond, it indicates a positive slope yield curve. According to risk structure of interest rate and term structure of interest rate, in general, the yield curve tends to have a positive slope. This is due to reasons such as default risk, liquidity, and preference for long-term bond. So, the YTM(yield to maturity) for long-term bonds be higher than YTM for short-term bonds. On the other hand, based on rational expectations theory, when the yield curve is positive slope, it implies that investors generally expect

short-term interest rates will rise in the future. As long-term bond rates increase, short-term bonds become more attractive, prompting investors to sell short-term bonds and buy long-term bonds. This increases the supply of short-term bonds, leading to a decrease in bond prices and an increase in short-term bond yields. Similarly, when the yield curve is negatively sloped, it suggests that investors generally anticipate a decline in short-term interest rates.

- (b) On December 3, 2018, the bond market witnessed an inverted yield curve phenomenon where the yield on 5-year bonds was lower than the yield on 3-year bonds. This occurred due to the Federal ongoing policy of raising interest rates and reducing its balance sheet. On one hand, this led to a continuous increase in short-term yields, even surpassing long-term yields. According to the rational expectations theory, when the yield curve inverts, it indicates the expectation of future declines in short-term yields. Anticipating a decrease in short-term yields is often seen as an early indicator of an economic downturn. On the other hand, based on historical experience, when the yield curve inverts, it is followed by an economic recession within a period of 6 months to a year. This is because the Fed's aggressive interest rate hikes can cool down the economy and even trigger expectations of a recession. Due to these reasons, when the yield curve inverts for the first time, it sparks concerns in the financial markets about an impending economic downturn, leading to a decline of approximately 800 points in the Dow Jones Index on that day.

解答太長了，給個中文版：

- (a) 當五年期債券殖利率高於二年期債券殖利率時，表示 yield curve 為正斜率。依據 risk structure 與 term structure 學說，正常情況下 yield curve 為正斜率，基於倒帳風險、流動性與期限偏好等原因，長期殖利率一般會高於短期殖利率。另一方面根據理性預期理論，當收益率曲線為正斜率時，表示投資人普遍預期未來短期利率會上升。因為當長期債券的利率上升時，會使短期債券變得更有吸引力，投資人會出售短期債券來購買長期債券，使短期債券供給增加，債券價格下跌，以至於短期債券的利率也跟著上升。同理，當收益率曲線為負斜率時，代表投資人普遍預期未來短期利率將會下跌。

- (b) 2018年12月3日債券市場出現五年期債券殖利率低於三年期債券殖利率，產生殖利率曲線倒掛的現象，此是由於 Fed 持續採取升息與縮表政策，一方面使短期殖利率不斷上漲，甚至高於長期殖利率，而依據理性預期理論，當殖利率曲線倒掛時，表示人們預期未來的短期殖利率將會回跌。而預期短期殖利率下跌恰好是景氣衰退的先行指標。另一方面依據過去的歷史經驗，當殖利率曲線逆轉時，在經過半年到一年後之期間內，將會帶來經濟衰退。這是因為 Fed 的強力升息將會引發經濟體的降溫，甚至發生衰退的預期。基於以上原因，當殖利率曲線第一次出現逆轉時，就會引起金融市場對經濟衰退的聯想，導致當天道瓊指數下跌了約800點。